Chengwang Liao

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Research Interests

Machine learning; high-dimensional econometrics; dynamic models.

Education

The Chinese University of Hong Kong — Hong Kong
Ph.D. in Economics — *2022–2027 (expected)*

Shanghai Jiao Tong University — Shanghai
Master of Economics — *2019–2022*

Huazhong University of Science and Technology — Wuhan
Bachelor of Economics — *2015–2019*

Job Market Paper

Recursive Shrinkage Local Projections
Solo-authored

Abstract. Local projections (LPs) estimate impulse response functions (IRFs) by running a series of regressions separately for each horizon. To utilize the nature restrictions across horizons, we propose a novel estimator called the recursive shrinkage local projections (RSLP). This estimator penalizes deviations from a linear recurrence of the underlying coefficients across horizons, obtaining the IRFs in a single programming problem. Under correct specification of the recurrence structure, we establish consistency and asymptotic normality, demonstrate efficiency gains over the vanilla local projections and develop statistical inference. The shrinkage strength is adaptively tuned to robustify the procedure even if the underlying recurrence is misspecified. Monte Carlo experiments show substantial estimation accuracy gains over the vanilla local projections. An application to monetary policy shocks yields smoother response paths and narrower confidence intervals.

Publications

Nickell Meets Stambaugh: A Tale of Two Biases in Panel Predictive Regressions
with Ziwei Mei and Zhentao Shi
Journal of Econometrics, 2026. Published online.

Working Papers

Cointegrated Regressors: Where Folded Concave Penalties Go beyond LASSO
with Hongqi Chen, Ziwei Mei, and Zhentao Shi
Submitted

A Relaxation Approach to Synthetic Control
with Zhentao Shi and Yapeng Zheng
Revise and resubmit, *Econometric Theory*, 2026

Conference and Seminar Presentations

2026 Brownbag Seminar, The Chinese University of Hong Kong
3rd Annual International Conference of the Greater Bay Econometrics Study Group, Sun Yat-sen University (Guangzhou)
Symposium on Artificial Intelligence and Financial Econometrics (Xiamen)

2025 19th International Symposium on Econometric Theory and Applications (Macau)
Harbin International Conference on Econometrics and Statistics (Harbin)
13th World Congress of the Econometric Society (Seoul)
6th Workshop on Big Data Econometrics: Theory and Applications (Wuhan)

2024 Brownbag Seminar, The Chinese University of Hong Kong
2nd Joint Workshop on Chinese Economy, The Chinese University of Hong Kong, Shenzhen and The Chinese University of Hong Kong

Teaching Experience

Teaching Assistant, The Chinese University of Hong Kong

Intermediate Macroeconomics	Fall 2025; Spring 2026
Basic Microeconomics	Spring 2025
International Monetary Systems	Fall 2024
Mathematical Methods in Economics	Spring 2023; Spring 2024
Introductory Econometrics	Fall 2022

Honors and Awards

First Class Academic Scholarship, Shanghai Jiao Tong University	2020–2021
National Scholarship, Huazhong University of Science and Technology	2018–2019

Skills

Languages Chinese (native), English (fluent)
Programming MATLAB, Python, R, LaTeX

References

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Junhui Qian, Professor
Antai College of Economics and Management, Shanghai Jiao Tong University
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